

HAMZA MAJID

- Germany • qazihamzamajid@gmail.com
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SUMMARY

AI/ML Engineer combining 2+ years of production software engineering in regulated financial environment and research at TU Berlin. Currently pursuing an MSc in Artificial Intelligence at Brandenburg University of Technology. Work includes a comparative study of Graph Neural Networks (GraphSAGE) and Transformers for road network analysis alongside time-series forecasting with LSTMs and NLP pipeline development. Skilled in Python, deep learning networks, and scalable system design. Seeking thesis and internship opportunities in AI engineering, autonomous intelligent network research in Germany.

WORK EXPERIENCE

Working Student | Technische Universität Berlin | Berlin, Deutschland | May 2026 – Present

- Deployed a software defined private 5G O-RAN testbed using OpenAirInterface and FlexRIC for network management and XAI validation.
- Built a real-time KPI pipeline (E2 → SQLite → InfluxDB → Grafana) capturing per-UE metrics for model training.
- Integrated RC service model and designed data export for RL-based resource allocation and optimization.
- Fixed critical E2 integration by replacing O-RAN SC RIC with FlexRIC, enabling stable end-to-end integration.
- Automated testbed setup and operations via Python and shell scripts for reproducibility.

Working Student | Technische Universität Berlin | Berlin, Deutschland | August 2025 – April 2026

- Optimized the Autonomous Reservoir Computing pipeline, increasing time-series prediction accuracy by 15%.
- Enhanced the PyReco open-source library (reservoir computing/ESN) used by AI/ML researchers.
- Developed an interactive neural network visualization tool, improving reservoir structure understanding for users.
- Implemented a robust error-handling framework using validator and decorator patterns, improving system reliability and user feedback.

Software Engineer | Bank Al Habib | Karachi, Pakistan | August 2022 – April 2025 |

(incl. Trainee, Aug 2022 – Mar 2023)

- Developed and maintained 9+ data-driven microservices using Java and Quarkus on Linux for financial operations.
- Improved system performance by reducing execution time by 40% through algorithm optimization and caching.
- Implemented Apache ActiveMQ based event-driven architecture, reducing inter-service latency by 30%.
- Designed IBM Db2 schemas and reporting solutions for financial analytics and risk monitoring.
- Contributed to agile development and code reviews, ensuring high-quality, production-ready code.

EDUCATION

Master of Science in Artificial Intelligence | April 2025 – Present

Brandenburg University of Technology, Cottbus

Bachelor of Science in Computer Science | September 2018 – May 2022

Institute of Business Management, Karachi

- **Quantitative & Financial Analysis:** Time-Series Analysis, Statistical Modeling, Regression, Forecasting, Optimization, Financial Data Analysis, Risk Metrics, Backtesting, Numerical Methods
- **Version Control & DevOps:** Git, GitHub, Jira, CI/CD, Docker
- **Tools & IDEs:** IntelliJ, VS Code, Eclipse; APIs: Postman/Insomnia
- **Languages:** English-C1, Urdu-Native, German-A2/B1

PROJECTS

Graph-Based Trajectory Prediction using GNNs and Transformers

- Built a road network graph from OpenStreetMap data (~2K nodes, ~5K edges) and generated synthetic trajectories via random walks.
- Implemented GraphSAGE for spatial relationships and a Transformer for sequential pattern modeling, training pipelines.
- Compared models on next-node prediction accuracy and cross-entropy loss: Transformer-based models outperformed GNNs on sequential trajectory modeling; GNNs captured spatial road structure more effectively. Results are directly applicable to autonomous driving architecture decisions.

Financial Time-Series Forecasting (Stock Market Prediction)

- Built a multi-ticker prediction pipeline for SPY, QQQ, and equities, comparing Logistic Regression, XGBoost, and LSTM using 55+ leakage-free technical and market-regime features.
- Implemented expanding-window walk-forward validation with purge/embargo gaps and transaction costs baseline (AUC ~0.50–0.54), plus back testing with transaction costs, Sharpe, drawdown, and position sizing overlays.
- Developed an interactive Streamlit dashboard for per-ticker training, walk-forward evaluation, testing, and predictions, with optional LSTM support and transparent performance reporting.

Recall — Multi-Document RAG Pipeline

- Built a multi-document RAG pipeline with vector database retrieval, comparing chunking strategies and embedding models on retrieval quality. Evaluated with recall @ k metrics against a held-out test set.